

has evolved over the years -- often under the influence of C++. When I started with 'C with Classes', the use of sets of functions as opaque interfaces (which today is C's most effective abstraction mechanism) was not common. It has been conjectured that this developed partly in response to the classes and virtual functions of C++. K&R C did not offer function prototypes, //, const, inline and more; those came from my work with C++. Even GCC (GNU Compiler Collection) that was first released in 1987 is a C++ program now.

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### **Q** How has C++ evolved in the programming space?

For the first 10 years or so, the user population of C++ doubled every 7.5 months. Currently, there are about 4.5 million users of C++, and that number is growing by about 100,000 a year.

Moreover, C++ is used all over the world, and heavily in finance, banking, games, front offices, telecom, electronics, marketing, manufacturing and retail. From my own experience, I can add embedded systems, scientific

computation and graphics to the list of use cases.

### **Q** Is it the open source practice that led to the early success for programming languages such as C and C++?

No. The early success of C and C++ predates the emergence of open source. However, AT&T did the next best thing and allowed the use of C and C++ compilers and libraries for a very low price. For non-profit organisations,



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creator of C++